Software Metrics & Complexity Analysis

Smart Waste Collection Management System

Note: weights and ratings below are estimates made from the SRS alone. Adjust them to match your course's method.

1. Requirements Metrics

Metric	Value
System features	5 (BM, AL, RO, DB, CF)
Functional requirements	15 (3 per feature)
Nonfunctional requirements	15 (performance 3, safety 2, security 3, quality 4, business rules 3)
Total requirements	30
FR : NFR ratio	1 : 1
High / Medium priority features	3 / 2
TBD items	3 (about 10% of requirements still open)

Specificity: Only about 7 of the 30 requirements have measurable acceptance criteria: 15-minute interval, 5-minute alert, 10 s for 200 bins, 500 nodes, 3 s load, 95% reliability, and HTTPS/TLS. That is about 23%. Phrases like “gracefully”, “securely”, “minimal training” and “real time” cannot be tested as written.

2. Function Point Analysis (IFPUG-style)

Type	Items	Weighted count
External Inputs (7)	Sensor data, threshold config, login, route request, route edit, complaint submit, mark collected	25
External Outputs (3)	Alert notification, optimized route, analytics report	16
External Inquiries (4)	Live bin map, bin history, complaint status, driver route view	14
Internal Logical Files (5)	Bins/readings, alert log, complaints, users, routes/collections	44
External Interface Files (1)	Mapping/routing API data	7
Unadjusted FP		**106**

Value Adjustment Factor: Rating the 14 General System Characteristics gives a total of 45 (high on data communications, performance, complex processing and online entry; low on reusability and installation ease). VAF = 0.65 + 0.01 × 45 = **1.10**.

Adjusted FP = 106 × 1.10 ≈ 117 FP

3. Effort Estimation (COCOMO)

Assuming about 50 LOC per FP for a JavaScript/Python stack, 117 FP ≈ **5.9 KLOC**.

Model	Effort	Duration	Avg. staff
Basic COCOMO, Organic ($E = 2.4 \cdot KLOC^{1.05}$)	≈ 15.3 person-months	≈ 7.0 months	≈ 2.2
Basic COCOMO, Semi-detached ($E = 3.0 \cdot KLOC^{1.12}$)	≈ 21.7 person-months	≈ 7.3 months	≈ 3.0

Semi-detached is probably the more realistic model because the project mixes hardware and software. A team of 4 over one semester is a reasonable fit. These figures come from a very rough LOC-per-FP conversion, so treat them as ballpark estimates.

4. Complexity Analysis

4.1 Cyclomatic complexity (estimated, V(G) = decisions + 1)

Module	Decision points	V(G)
Sensor read and fill %	read failed? retry limit?	~3
Alert evaluation	valid reading? ≥ threshold? duplicate alert?	~4
Route generation	filter bins ≥ threshold, empty set, loop over bins, manual add/remove	~6–8
Complaint auto-resolve	bin collected? complaint pending?	~3
Dashboard auth	valid credentials? OTP enabled? OTP valid?	~4

All modules are in the low-risk range (V(G) < 10). Route generation is the most complex module and the one to unit-test most heavily.

4.2 Algorithmic complexity (route optimization, TBD-1)

- Nearest-Neighbor heuristic is $O(n^2)$. For n = 200 that is about 40,000 operations, far below the 10-second limit in REQ-RO-2.
- An exact Vehicle Routing Problem solver is NP-hard. It would need a heuristic or metaheuristic (for example, OR-Tools) to meet the time limit.
- For a prototype, Nearest-Neighbor plus 2-opt improvement is a defensible choice.

4.3 Data and load estimates

- 500 nodes × 96 readings/day (every 15 min) = **48,000 readings/day**, or about 17.5 million per year.
- Average ingestion is about 0.56 messages/s (500 ÷ 900 s), easy for either MongoDB or PostgreSQL. Storage and archiving policy matter more than throughput.

4.4 Architectural metrics (qualitative)

- 5 modules (monitoring, alerts, routing, dashboard, complaints) and 4 deployable components (sensor node, backend, web, mobile).
- Cohesion looks high because each feature maps to one module. Coupling is moderate: alerts depend on monitoring, routing depends on both, and complaint resolution depends on the driver's "collected" event.

5. Issues the Metrics Exposed

1. **Timing conflict:** REQ-AL-2 requires an alert within 5 minutes of a bin crossing the threshold, but readings arrive every 15 minutes. It should say "within 5 minutes of the reading being received."
2. **Missing requirement:** The business rule auto-resolves complaints when a bin is "collected", and the driver UI has a "mark collected" action, but no REQ-xx covers it.
3. **Ambiguous thresholds:** The "alert threshold" and "collection threshold" are used as if they were the same value. Decide whether they are one setting or two.
4. **"Real-time" wording:** Section 4.1.2 says "real time", but data refreshes only every 15 minutes.
5. **Missing web form requirement:** Section 2.3 mentions a citizen web form, but no requirement covers it.
6. **Open items:** Appendix B (diagrams) is empty and the three TBDs are unresolved, so the SRS is not fully complete.